On a question about weak semantic conservativeness

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Abstract

We settle a question about weak semantic conservativeness raised by Stephen Mackereth.

Neologicists (amongst others) have explored a number of ways of articulating precisely a *conservativeness* condition on stipulative definitions, of which Hume's Principle is often taken to be an example.[1] In [2], Mackereth usefully distinguishes between three notions: *deductive*, *strong semantic*, and *weak semantic* conservativeness. Our aim in this note is to settle a question of Mackereth's concerning the third of these notions of conservativeness.

To this end, let us first recall the definition of weak semantic conservativeness given in [2].

Definition (Weak semantic conservativeness)**.** Let T be a theory in a language $\mathcal{L}$, and let Δ be a set of sentences in a language $\mathcal{L}' \supset \mathcal{L}$. Then Δ is *weakly semantically conservative over* T iff every $\mathcal{L}$-formula that is a consequence of $T \cup \Delta$ is already a consequence of T alone.

Note that we are concerned here with consequence in *full* (as opposed to Henkin) second-order semantics: that is, we shall be concerned only with models $\mathcal{M}$ in which n-adic second-order variables range over the full power-set $\mathcal{P}(D^n)$, where D is the domain (of individuals) of $\mathcal{M}$.

We can now state the question posed by Mackereth in [2, p. 14]:

Problem. *If* Δ_1 *and* Δ_2 *are each weakly semantically conservative over* T *(in SOL), and introduce disjoint sets of new vocabulary, does it follow that they are* jointly *weakly semantically conservative over* T*?*

[1] See, for example, Hale and Wright [1]. However, Mackereth [2, fn. 12] notes that in more recent writings, Wright may have moved away from the claim that Hume's Principle is a stipulative definition (of cardinal number).

In general (that is, for arbitrary $T, \mathcal{L}$), the answer to this question turns out to be negative. More precisely:

Theorem 1. *There is a language $\mathcal{L}$ (extending the language of pure SOL, $\mathcal{L}_{\mathrm{SOL}}$) and an $\mathcal{L}$-theory T, and sets of sentences Δ_1, Δ_2 which introduce disjoint sets of new vocabulary to $\mathcal{L}$, such that Δ_1 and Δ_2 are individually but not jointly weakly semantically conservative over T.*

Proof. Let $\mathcal{L}$ be $\mathcal{L}_{\mathrm{SOL}}$ augmented by a countable infinity of propositional constants $\{p_n : n \in \mathbb{N}\}$,[2] and let T consist of a single sentence stating that the universe is countably infinite. Then every model $\mathcal{M}$ of T is determined (up to isomorphism) by the function $\mathrm{val}_{\mathcal{M}} : \mathbb{N} \to \{0, 1\}$ which gives the truth-value of each p_n in $\mathcal{M}$.

Now let $\mathcal{L}_1 = \mathcal{L} \cup \{N_1, Z_1, X_1, Q_1\}$ and $\mathcal{L}_2 = \mathcal{L} \cup \{N_2, Z_2, X_2, Q_2\}$, where N_1, N_2, Q_1, Q_2 are unary predicates, Z_1, Z_2 are individual constants, and X_1, X_2 are unary function symbols, all mutually distinct. We take Δ_1 to be the set of $\mathcal{L}_1$-sentences

$$\{\text{`}(N_1, Z_1, X_1) \text{ is isomorphic to } (\mathbb{N}, 0, S)\text{'}\} \cup \{Q_1(X_1^n(Z_1)) \leftrightarrow p_n : n \in \mathbb{N}\} \cup \{\text{`the extension of } Q_1 \text{ is finite'}\}$$

and similarly take Δ_2 to be the set of $\mathcal{L}_2$-sentences

$$\{\text{`}(N_2, Z_2, X_2) \text{ is isomorphic to } (\mathbb{N}, 0, S)\text{'}\} \cup \{Q_2(X_2^n(Z_2)) \leftrightarrow \neg p_n : n \in \mathbb{N}\} \cup \{\text{`the extension of } Q_2 \text{ is finite'}\}.$$[3]

Thus $\mathcal{M}$ expands to a model of $T \cup \Delta_1$ (resp. $T \cup \Delta_2$) iff $\mathrm{val}_{\mathcal{M}}$ is eventually 0 (resp. eventually 1). Since no function $\mathbb{N} \to \{0, 1\}$ can be both eventually 0 and eventually 1, we find that $T \cup \Delta_1 \cup \Delta_2$ has no model, and therefore that $\Delta_1 \cup \Delta_2$ is not weakly semantically conservative over T.

To see that, on the other hand, each of Δ_1 and Δ_2 *is* weakly semantically conservative over T, suppose that $\mathcal{M}$ is a model of $T \cup \{\neg\phi\}$, where no p_n occurs in ϕ with $n \geq k$. Then letting val agree with $\mathrm{val}_{\mathcal{M}}$ on $\{0, \ldots, k-1\}$ and setting $\mathrm{val}(n) = 0$ (resp. $\mathrm{val}(n) = 1$) for $n \geq k$ yields a model of Δ_1 (resp. Δ_2) which (still) satisfies $T \cup \{\neg\phi\}$, as required. □

In contrast to this result, we can give an affirmative answer to Mackereth's question for the special case where T is empty, and what is at issue is weak semantic conservativeness *over pure SOL*.

[2] We could instead work with a countable infinity of unary predicates P_n, and let T in addition say of (the extension of) each predicate that it is either empty or universal.

[3] Where, as usual, $X_i^0(Z_i) := Z_i$ and $X_i^{n+1}(Z_i) := X_i(X_i^n(Z_i))$, for $i = 1, 2$.

Theorem 2. *If Δ_1 and Δ_2 are each weakly semantically conservative over pure SOL and introduce disjoint sets of new vocabulary, then so is $\Delta_1 \cup \Delta_2$.*

Proof. We first observe that if Δ is weakly semantically conservative over pure SOL and σ is a categorical sentence of $\mathcal{L}_{\mathrm{SOL}}$, that is, has exactly one model $\mathcal{M}$ (up to isomorphism), then $\mathcal{M}$ expands to a model of Δ. For given any model of $\Delta \cup \{\sigma\}$ (there must be at least one: since $\nvDash \neg\sigma$, conservativeness assures us that $\Delta \nvDash \neg\sigma$), its $\mathcal{L}_{\mathrm{SOL}}$-reduct is isomorphic to $\mathcal{M}$, and we simply push the expansion through the isomorphism.

Suppose that ψ is satisfiable. Then there is a categorical sentence σ such that $\{\sigma\} \vDash \psi$: for example, take $\sigma = \psi \wedge \forall X\big((\exists x Xx \wedge \neg X \approx U) \rightarrow \neg\psi^X\big)$,[4] where U denotes the universal concept ($Ux \leftrightarrow x = x$) and ψ^X is ψ with all (first-order) quantifiers relativised to X. (The models of σ are then exactly the models of ψ of least cardinality, any two of which have equinumerous domains and so are isomorphic as models of pure SOL.) By the above observation, any model $\mathcal{M}$ of σ (and so of ψ) expands to a model of Δ_1 and a model of Δ_2; and since the (new) vocabularies introduced are disjoint, the expansions can be combined to give a model of $\Delta_1 \cup \Delta_2 \cup \{\sigma\}$. □

We note in passing that this result holds also for any finitely axiomatised T in pure SOL: in the above argument, simply replace ψ by $\psi \wedge \tau$, where τ is the conjunction of the axioms of T.

In light of Mackereth's discussion in [2, sect. 4], the interest of Theorem 2 to the neologicist will be obvious. If the stipulative definitions D_1 and D_2 introduce disjoint sets of vocabulary and are each weakly semantically conservative over pure SOL, then they will be jointly weakly semantically conservative and therefore jointly consistent. Thus requiring that stipulative definitions be weakly semantically conservative (over pure SOL) suffices to rule out (strong) Bad Company.[5]

References

[1] B. Hale and C. Wright (2001), *The Reason's Proper Study: Essays Towards a Neo-Fregean Philosophy of Mathematics*, Oxford: Clarendon Press.

[2] S. Mackereth (2026), Neo-logicism and conservativeness, *Journal of Philosophy* 123(1): 5–34.

[4] '$X \approx Y$' abbreviates 'there is a bijection between X and Y'.

[5] Thanks to Claude (Opus 5.5) for assistance with proofreading, and for the observation immediately following Theorem 2.